MOHAMMED HADI

Manipal, India | +91 9353766821 | mhdhadiabdullah@gmail.com | github.com/mhdhadi2005 | LinkedIn | mohammed-hadi.vercel.app

RESEARCH INTERESTS

Machine learning for return prediction and empirical asset pricing, with an emphasis on honest out-of-sample evaluation (leakage, purged walk-forward validation, naive baselines, transaction costs) · market microstructure and high-frequency data · decentralized finance and crypto-asset markets · the data and systems infrastructure behind trading research.

EDUCATION

Manipal Institute of Technology (MIT Manipal), Manipal Academy of Higher Education *07/2023 – Present*
Bachelor of Technology, Data Science and Engineering — Manipal, India

FINANCE RESEARCH PROJECTS

QuantAI — Do ML models have a real edge in financial time series? *— Python, Pandas, Scikit-learn, XGBoost, LSTM, FastAPI*

- Built an XGBoost + LSTM ensemble that predicts 5-day price direction from 36 technical-indicator features, served through FastAPI, with a rule-based backtester (ATR position sizing, stop-loss/take-profit, commission and slippage).
- Audited its evaluation and found it inflated: the model-driven backtest trained on its own test window, and accuracy was never compared with a naive baseline.
- Tested it on 50 randomly drawn S&P 500 stocks (2010–2026), changing one evaluation choice at a time: XGBoost's edge over an always-up baseline was +39 pp when scored on its training window but −6 pp on held-out data, and under a purged walk-forward no model (logistic, XGBoost, properly trained LSTM) beat the baseline. Working paper: *Where Did the Edge Go?*

In-browser backtester on live crypto data *— TypeScript, CoinGecko API*

- SMA-crossover backtest on real BTC 4-hour candles that runs in the browser: signals form on bar *t*'s close and fill on bar *t+1*, with trading fees included, so results reflect executable trades.

Competitive trading

- Placed 5th nationally in the Big Bull Trading Competition (Master's Union x CaseUnion).

DATA, ML & SYSTEMS PROJECTS

redis-java — Replicated key-value server from scratch *— Java, TCP, Docker, Kubernetes/Helm*

- Implemented the Redis RESP wire protocol and a concurrent multi-client TCP server (SET with PX expiry, GET, INCR, DEL, INFO), debugging concurrency edge cases; added master/replica replication and a documented Helm chart.

ai-cookbook — Retrieval and LLM-agent patterns *— Python, LLM APIs, MCP*

- Documented hybrid retrieval (BM25 and dense retrieval fused with Reciprocal Rank Fusion), agentic RAG, structured-output workflows and a Model Context Protocol crash course. Relevant to text-as-data methods for financial documents.

minic — C compiler in C *— C, x86-64, System V ABI*

- Compiles a subset of C (pointers, arrays, control flow, functions) to x86-64 assembly that links against libc.

EXPERIENCE

Web Development Intern, Montagu Publishing *(Remote)* *01/2026 – Present*

- Built and run a single-variable A/B testing library for lead-magnet pages (exit-intent popups, sticky CTAs, social proof), designing each experiment from stakeholder feedback against funnel goals for a US audience.
- Ship landing pages across newsletter brands (WordPress, Elementor, Beehiiv) and fixed a recurring nested-container layout bug with a reusable fix.

Regional Head for Business Development, AIESEC *(Manipal, India)* *01/2025 – 12/2025*

- Led a business-development team and negotiated B2B and sponsorship partnerships, using data-driven insight to hit national and regional targets.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL · **Quant / ML:** Pandas, NumPy, Scikit-learn, XGBoost, LSTM, backtesting, time-series validation, LLM APIs, RAG · **Infra:** FastAPI, Redis, MongoDB, MySQL, Docker, Kubernetes (Helm), GitHub Actions, Azure, GCP, AWS

AWARDS & CERTIFICATIONS

5th (National), Big Bull Trading Competition — Master's Union x CaseUnion | Top 3, Sharjah Chamber Award for Innovators | Microsoft Azure for Data Engineering | Python for Data Science, AI & Development — IBM